



Book I. Prop. XXXII. Theor. *If any side (—) of a triangle be produced, the external angle (◐) is equal to the sum of the two internal and opposite angles (◑ and ◒), and the three internal angles of any triangle taken together are equal to two right angles.*

Through the point  draw
 $\parallel$  (pr. I.31).

Then $\left\{ \begin{array}{l} \text{◐} = \text{◑} \\ \text{◑} = \text{◒} \end{array} \right\}$ (pr. I.29),

$\therefore \text{◑} + \text{◒} = \text{◐}$ (ax. II),

and therefore

$\text{◑} + \text{◒} + \text{◓} = \text{◐} + \text{◓} = \text{◑} + \text{◒} + \text{◓}$ (pr. I.13).

Q. E. D.